The Water Cycle

Water on the Move

Earth's water is always travelling. It lifts off the oceans as invisible vapour, gathers into clouds, falls back to the ground as rain or snow, and then runs downhill to begin the journey again. Scientists call this endless loop the **water cycle**.

The cycle has no true beginning. Most textbooks start with evaporation anyway, because the sun's heat is what keeps the whole system running. Without that energy the air would hold almost no moisture, and the continents would slowly dry out.

Water spends wildly different amounts of time in each stage. A droplet may hang in the atmosphere for roughly nine days, while the same molecule can sit inside a glacier for thousands of years before it melts and moves on again.

A Voice from the Mountains

In every walk with nature one receives far more than he seeks.

— John Muir

Muir was writing about mountain trails, but the point carries over to water. Paying close attention to something as ordinary as rainfall rewards the person who bothers to look.

Words Worth Knowing

Evaporation is the change from liquid water to vapour. *Transpiration* is the matching process in plants, where moisture escapes through tiny leaf pores called stomata. Put the two together and you get evapotranspiration — a clumsy word for a simple idea: water leaving the land and heading skyward.

Try It Yourself

Fill a shallow dish with warm water and set it on a sunny windowsill. Mark the water level with a pencil, then check it again the next day. The drop you measure is the first stage of the cycle, happening in your own kitchen.